

Camera

m_EyePoint
m_xVector
m_yVector
m_zVector
m_Fov
m_DistanceToObject
m_FocalLength
m_ScreenWidth
m_ScreenHeight
m_aspectRatio
m_LastLookAtPoint

+ Camera()
+ setEyePoint()
+ setDistanceToObject()
+ setFocalLength()
+ lookAt()
+ moveInGlobalFrame()
+ moveInLocalFrame()
+ zoomCamera()
+ moveInGlobalFrameFixLookAt()
+ moveInLocalFrameFixLookAt()
[5 0 B] [4 3 U B] 14 [4 5 P 6]...

QOpenGLWidget

Object

+ meshes

MyOpenGLWidget

width
height
g_FrameSize_WindowSize_Scale_x
g_FrameSize_WindowSize_Scale_y
orthoMode
g_AreaLights_preview
m_objFileName
- lastMousePos
- desiredAspectRatio
- button
- zoombutton
- lastButtonPos
- ButtonPressed
- lastZoomButtonPos
- zoomButtonPressed
- originalMousePos
- cameraFov

+ MyOpenGLWidget()
+ setCameraFov()
+ setOrthoMode()
+ getOrthoMode()
+ getEyePoint()
+ getlookAtPoint()
+ getUpVector()
+ getXVector()
+ getYVector()
+ getZVector()
[5 0 B] [4 3 U B] 7 [4 5 P 6]...
initializeGL()
resizeGL()
updateProjectionMatrix()
paintGL()
mousePressEvent()
mouseMoveEvent()
mouseReleaseEvent()
wheelEvent()
eventFilter()
projection_and_modelview()
initializeButtons()
initAreaLights()
onButtonPressed()
onButtonReleased()
onButtonMoved()
onZoomButtonPressed()
onZoomButtonReleased()
onZoomButtonMoved()

FilmBuffer

- m_width
- m_height
- m_filmBuffer
- m_accumulationBuffer
- m_countBuffer

+ FilmBuffer()
+ ~FilmBuffer()
+ init()
+ reset()
+ addSample()
+ updateFilmBuffer()
+ getFilmBufferPtr()
+ getWidth()
+ getHeight()

MyOpenGLWidget_camera

m_timer
m_isDirty
m_progress_i
m_progress_j
- vertex
- face
- color
- updatedFov
- m_filmTexture
- g_AreaLights

+ MyOpenGLWidget_camera()
+ setCamerakeyframe()
+ setCameraEyePoint2()
+ setlookAtPoint2()
+ onCameraFovChanged()
initializeGL()
paintGL()
resizeGL()
mouseMoveEvent()
mousePressEvent()
mouseReleaseEvent()
wheelEvent()
updateProjectionMatrix()
saveCameraState()
loadCameraState()
resetRendering()
- drawCube()
- initializeFilmTexture()
- createDebugPattern()
- debug_computeNormalColor()
- updateRayTracing()

-g_Camera

#g_Obj_preview

-g_Camera2

-g_Obj

-m_film